Euclid, Book I, Proposition I.13

If a Straight Line Standing on a Straight Line Makes Angles, It Makes Either Two Right Angles or Angles Equal to Two Right Angles

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1 Introduction

Euclid's Proposition I.13 concerns the two adjacent angles formed when one straight line stands on another straight line.

Let C, B, D lie on one straight line, with B between C and D, and let A be a point outside that line. The two adjacent angles are

$$\angle CBA \qquad \text{and} \qquad \angle ABD.$$

Euclid states that these angles are either two right angles or are together equal to two right angles.

The classical proof distinguishes two cases. If the adjacent angles are equal, then by Definition 10 they are both right angles. If they are not equal, Euclid erects a perpendicular at B using Proposition I.11 and compares the resulting angle decompositions.

The reconstruction over the Hilbert/Book Zero layer is much shorter. Book Zero already contains an explicit relation expressing that one angle is supplementary to another:

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BookZeroSupplement
```

For the configuration of Proposition I.13, this relation contains exactly the required geometric data. Consequently, the formal theorem does not reconstruct Euclid's case distinction or introduce an arithmetic operation on angles. It identifies the two adjacent angles directly as a supplementary pair.

Thus Proposition I.13 provides a particularly clear example of the compression obtained from the Book Zero language: a classical geometric argument becomes essentially an instance of an already available synthetic relation.

2 The Classical Configuration

Proposition I.13 is not a construction theorem. The geometric configuration is already given.

Let C, B, D lie on one straight line with

$$C - B - D,$$

and let A lie outside the straight line CD.

The straight line BA stands on the straight line CD as shown in Figure 1 and forms the adjacent angles

$$\angle CBA \qquad \text{and} \qquad \angle ABD.$$

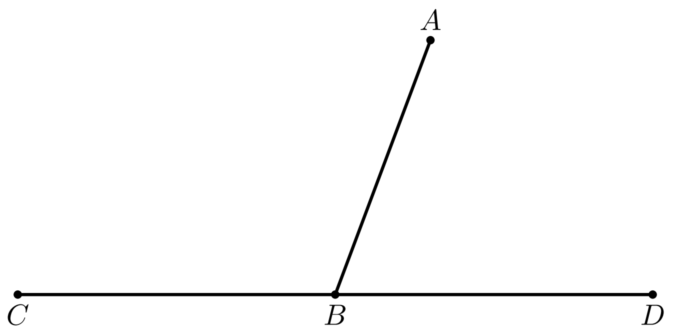

Figure 1: Classical configuration for Proposition I.13. The relation $C - B - D$ makes BC and BD opposite rays, while BA is the standing ray.

3 Statement

Assume

$$C - B - D$$

and let A be a point distinct from B.

Then the adjacent angles

$$\angle CBA \qquad \text{and} \qquad \angle ABD$$

form a supplementary pair.

In Euclid's formulation, they are therefore either two right angles or together equal to two right angles.

In the Book Zero language this conclusion is represented by

$$\text{BookZeroSupplement}(C, B, A, A, D).$$

By definition,

$$\text{BookZeroSupplement}(C, B, A, A, D)$$

means

$$\text{SameRay}(B, A, A)$$

together with

$$C - B - D.$$

The first condition expresses the trivial fact that the ray BA coincides with itself, while the second is precisely the given straight-line configuration.

4 Classical Proof

Euclid proves Proposition I.13 by distinguishing two cases.

4.1 Case 1. The Adjacent Angles Are Equal

Suppose first that

$$\angle CBA \cong \angle ABD.$$

Since C, B, D lie on one straight line, the two angles are adjacent angles formed by the straight line BA standing on CD.

By Euclid's Definition 10, when a straight line standing on another straight line makes the adjacent angles equal, each of those angles is a right angle.

Therefore

$$\angle CBA \qquad \text{and} \qquad \angle ABD$$

are two right angles.

This is the first alternative in Proposition I.13.

4.2 Case 2. The Adjacent Angles Are Not Equal

Suppose now that

$$\angle CBA \ncong \angle ABD.$$

By Proposition I.11, construct through B a perpendicular BE to the straight line CD; see Figure 2.

Thus

$$\angle CBE \qquad \text{and} \qquad \angle EBD$$

are right angles.

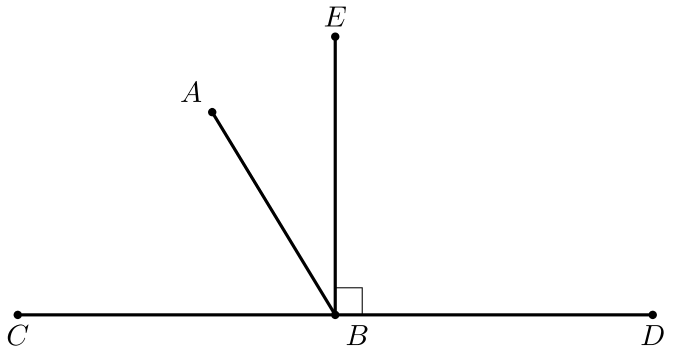

Figure 2: Auxiliary perpendicular configuration in the non-equal-angle case. The ray BE is perpendicular to CD, and BA lies between the rays BC and BE in the displayed arrangement.

Since BE is perpendicular to CD,

$$\angle CBE + \angle EBD$$

is equal to two right angles.

The original ray BA divides one of these right-angle configurations. In the arrangement shown in Figure 2,

$$\angle CBE = \angle CBA + \angle ABE.$$

Adding $\angle EBD$ gives

$$\angle CBE + \angle EBD = \angle CBA + \angle ABE + \angle EBD.$$

But

$$\angle ABE + \angle EBD = \angle ABD.$$

Hence

$$\angle CBE + \angle EBD = \angle CBA + \angle ABD.$$

The left-hand side consists of two right angles. Therefore

$$\angle CBA + \angle ABD$$

is equal to two right angles.

This is the second alternative in Proposition I.13.

Only the two numbered figures above are needed. Figure 1 states the given geometry; Figure 2 introduces the one genuinely new construction. Additional copies of the same rays would not correspond to new geometric information.

Thus, whether the two original adjacent angles are equal or unequal, they are either themselves two right angles or are together equal to two right angles.

5 Proof Architecture of the Reconstruction

Euclid's proof proceeds through the geometry of right angles and angle addition:

```
Euclid
------
C-B-D and standing ray BA
        |
are the adjacent angles equal?
        |
   +----+----+
   |         |
  yes        no
   |         |
Definition 10  erect perpendicular BE
   |         |
right angles  angle decomposition
   |         |
   +----+----+
        |
two right angles
```

```
Hilbert / Book Zero
-------------------
C-B-D
  +
SameRay(B,A,A)
        |
        v
BookZeroSupplement(C,B,A,A,D)
        |
        v
euclid_proposition_13
```

The two architectures should not be represented by additional geometric figures. The difference is logical rather than spatial: Euclid proves a statement about angle sums, whereas the Book Zero theorem recognizes the same straight-angle configuration directly.

The contrast is substantial. Euclid proves the proposition by introducing a perpendicular and reasoning about decompositions and combinations of angles. At the Book Zero level, the same geometric configuration has already been abstracted as the supplementary-angle relation.

No new geometric construction is therefore required, and Proposition I.11 disappears from the formal dependency path.

6 Hilbert Reconstruction

The Hilbert reconstruction of Proposition I.13 does not reproduce Euclid's proof.

The essential observation is that the conclusion of the proposition has already been isolated in Book Zero as the relation

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BookZeroSupplement
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For five points A, B, C, D, F, this relation is defined by

$$\text{BookZeroSupplement}(A, B, C, D, F)$$

if and only if

$$\text{SameRay}(B, C, D)$$

and

$$A - B - F.$$

Thus the relation records synthetically the configuration in which one side of an angle is continued through its vertex while the other side is preserved along the same ray.

For Proposition I.13 we instantiate this relation as

$$\mathrm{BookZeroSupplement}(C, B, A, A, D).$$

Expanding the definition gives exactly two conditions:

$$\mathrm{SameRay}(B, A, A)$$

and

$$C - B - D.$$

The second condition is one of the hypotheses of Proposition I.13.

The first condition says only that the ray BA is the same ray as itself. Since $A \neq B$, it follows immediately from reflexivity of the same-ray relation.

Consequently,

$$\mathrm{BookZeroSupplement}(C, B, A, A, D)$$

follows directly.

No perpendicular is constructed. No case distinction between equal and unequal adjacent angles is required. No decomposition or addition of angles occurs in the proof.

This does not mean that Euclid's geometric argument has been discarded. Rather, the geometric notion expressed by that argument has already been incorporated into the language of Book Zero. Proposition I.13 therefore becomes a recognition theorem: the given configuration is an instance of an existing synthetic relation.

7 Lean Formalization

The Lean theorem is correspondingly short:

```
theorem euclid_proposition_13
[HilbertIncidence Geo]
[HilbertOrder Geo]
(C B D A : Geo.Point)
(hCBD : Geo.Between C B D)
(hBA : B != A) :
BookZeroSupplement Geo C B A A D := by

constructor

· exact
hilbert_sameRay_refl
Geo B A hBA.symm

· exact hCBD
```

The proof follows the definition of

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BookZeroSupplement
```

literally.

The first component of the conclusion is

$$\mathrm{SameRay}(B, A, A).$$

It is obtained from

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hilbert_sameRay_refl
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using $A \neq B$, which is supplied by the symmetric form of the hypothesis $B \neq A$.

The second component is

$$C - B - D,$$

which is exactly the hypothesis

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hCBD
```

Thus the formal proof contains no intermediate geometric construction and requires no theorem corresponding to Euclid's use of Proposition I.11.

8 Discussion

Proposition I.13 is a particularly clear test of the role assigned to Book Zero in the reconstruction.

In Euclid's original development, the statement requires an actual proof. The argument distinguishes whether the two adjacent angles are equal. In the nontrivial case, Proposition I.11 is invoked to erect a perpendicular, and the conclusion is obtained by decomposing and recombining angles.

At the Hilbert level, one could reconstruct this argument explicitly. The existing theory already contains the necessary machinery for right angles, supplementary angles, and angle addition.

That reconstruction is unnecessary, however, once the Book Zero language is used.

The relation

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BookZeroSupplement
```

was introduced independently as a reusable description of a standard synthetic configuration. Proposition I.13 is precisely an instance of that configuration.

The resulting compression is therefore structural rather than tactical. Lean is not discovering a shorter proof of Euclid's argument. The formal language has been organized at a level at which the conclusion of I.13 can already be stated directly.

This produces an unusually short dependency path:

$$\text{betweenness} \quad + \quad \text{same-ray reflexivity} \quad \longrightarrow \quad \text{BookZeroSupplement} \quad \longrightarrow \quad \text{Euclid I.13.}$$

In particular, Proposition I.11, although essential to Euclid's classical proof, is not a dependency of the reconstructed theorem.

This distinction is important. The objective of the reconstruction is not to force every Euclidean proposition to retain the dependency graph of the original text. The objective is to express Euclidean geometry over a stable synthetic interface derived from the Hilbert foundation.

Earlier propositions exhibited several possible relationships with this interface. Some consumed already available Book Zero results, while others exposed missing reusable geometry that had to be added to the interface.

Proposition I.13 exhibits a third and especially simple situation: the proposition is already expressed by the vocabulary of Book Zero.

Schematically,

$$\begin{array}{ccl} \text{Euclid} & : & \text{case distinction} + \text{I.11} + \text{angle composition} \\ \text{Hilbert/Book Zero} & : & \text{recognition of a supplementary configuration} \\ \text{Lean} & : & \text{same-ray reflexivity} + \text{given betweenness} \end{array}$$

The four-line proof is therefore not merely an accidental shortening of Proposition I.13. It is evidence that the Book Zero layer is functioning as intended: elementary synthetic arguments can be encapsulated once and subsequently used as part of the language in which later geometry is formulated.

9 Relationship with Earlier Propositions

9.1 Proposition I.11

Proposition I.11 constructs a perpendicular to a given straight line at a given point on it. In the classical development this supplies the canonical right-angle configuration against which the adjacent angles in I.13 can be compared.

The formal reconstruction can package the same geometry through the Hilbert and Book Zero angle machinery, but the conceptual role remains the same: a perpendicular provides the reference configuration for two right angles.

9.2 Proposition I.12

Proposition I.12 extends the perpendicular construction to a point outside a line. It is not the principal dependency of I.13, but together with I.11 it completes the elementary perpendicular infrastructure that precedes the straight-angle block.

9.3 Transition to Proposition I.14

Proposition I.13 is the direct theorem: when a straight line stands on another straight line, the adjacent angles are two right angles or are together equal to two right angles.

Proposition I.14 reverses this logic. From the corresponding supplementary-angle condition it recovers the straight-line configuration.

Thus I.13 and I.14 form a natural theorem/converse pair and should be read as one local unit in the development of straight-angle geometry.

10 Hilbert and Book Zero Components Used

The reconstruction of Proposition I.13 relies on the Book Zero relation

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BookZeroSupplement
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defined by

$$\mathrm{BookZeroSupplement}(A, B, C, D, F)$$

if and only if

$$\mathrm{SameRay}(B, C, D)$$

and

$$A - B - F.$$

Geometrically, this represents a pair of supplementary angles. The angle determined by the rays BA and BC has as its supplement the angle determined by the rays BD and BF, where BD lies on the same ray as BC and BF is the opposite extension of BA.

For Proposition I.13 the relation is instantiated as

$$\mathrm{BookZeroSupplement}(C, B, A, A, D).$$

Thus the required conditions become

$$\mathrm{SameRay}(B, A, A)$$

and

$$C - B - D.$$

The first is immediate from same-ray reflexivity, while the second is the given betweenness hypothesis.

The proof also uses

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hilbert_sameRay_refl
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which expresses reflexivity of a nondegenerate ray:

$$A \neq B \quad \Longrightarrow \quad \mathrm{SameRay}(A, B, B).$$

11 Formalization Notes

The formal reconstruction is considerably shorter than Euclid's classical proof because the supplementary-angle configuration has already been packaged in Book Zero.

The theorem does not reproduce Euclid's case distinction and does not construct the perpendicular from Proposition I.11. Instead, it proves

$$\mathrm{BookZeroSupplement}(C, B, A, A, D)$$

directly from the two components required by the definition:

$$\mathrm{SameRay}(B, A, A) \qquad \text{and} \qquad C - B - D.$$

The first component follows from

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\thilbert_sameRay_refl
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using the nondegeneracy hypothesis $A \neq B$. The second component is exactly the given betweenness hypothesis.

Thus no numerical angle measure, angle addition, right-angle case analysis, or auxiliary perpendicular is needed in the Lean proof. The formalization records the same synthetic content at a more abstract level: Proposition I.13 is recognized as an instance of the already established Book Zero notion of supplementary angles.

This is also the form needed by Proposition I.14, which uses the same supplementary-angle structure in the converse direction.

12 Dependency Summary

The formal dependency structure is

a straight line standing on a straight line
$\downarrow$
betweenness / collinearity data
$\downarrow$
adjacent-angle configuration
$\downarrow$
comparison with a right-angle configuration
$\downarrow$
equal-angle or unequal-angle branch
$\downarrow$
two right angles or supplementary angles
$\downarrow$
`euclid_proposition_13`.

The proof uses only the established Hilbert and Book Zero infrastructure. No new geometric axiom is introduced.

13 Conclusion

Proposition I.13 begins the straight-angle block of Book I.

Its familiar classical statement looks almost arithmetical: the two adjacent angles made by a line standing on another line amount to two right angles. The formal reconstruction shows that no numerical angle measure is required to express or prove this fact.

Instead, the theorem is recovered from explicit incidence, order, right-angle, and angle-congruence structure. The classical distinction between equal and unequal adjacent angles is preserved, while the diagrammatic assumptions concerning the supporting straight line are made explicit.

The proposition is also structurally important. I.14 supplies its converse, and I.15 completes the local theory by proving the congruence of vertical angles. I.13 therefore marks the beginning of a coherent three-proposition development of the geometry of intersecting straight lines.